

Marker Signal Void: Linguistic Deception Fingerprinting and Transformer Benchmarking on a Domain-Stratified Bangla Rumour Corpus

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Abstract

Rumour detection in Bangla remains severely under-resourced, with prior work conflating it with fake news classification and lacking domain-stratified analysis of the linguistic properties that distinguish misinformation from authentic content. We introduce MisInfo-11K, a domain-stratified corpus of 11,198 instances spanning seven topical domains, constructed from Bangla fact-checking portals and verified news outlets. Using 14 linguistically motivated markers, we show via chi-square and Kruskal-Wallis analysis that 13 of 14 exhibit significant cross-domain variation, revealing distinct deception strategies (e.g., authority mimicry in political rumours, fear amplification in health rumours). Building on this, we introduce the Marker Signal Void (MSV), a chi-square-weighted scoring framework identifying rumour instances with no net discriminative surface signal, establishing Macro-F1 = 0.9843 as the ceiling achievable by this 14-marker set. Benchmarking three pre-trained transformers (BanglaBERT, MuRIL, XLM-RoBERTa) against five traditional models, MuRIL achieves the strongest performance (Macro-F1 0.9800), while a marker-augmented Random Forest reaches 0.9750, surpassing XLM-RoBERTa. The gap between the MSV ceiling and transformer performance quantifies the contribution of contextual semantic representations beyond surface-level linguistic cues.

1 Introduction

The 2019 Padma Bridge lynchings in which crowds killed several people acting on a false Facebook rumour that children’s heads were needed as sacrificial offerings for the bridge’s construction illustrate that unverified claims in active circulation cause physical harm well before their veracity can be established (Shirina and Prodhan, 2020). Yet computational approaches to Bangla *rumour detection* remain absent from the literature: existing work addresses fake news detection binary

classification of fabricated articles which is categorically distinct from rumour detection, since rumours are unverified claims whose veracity is actively contested at propagation time (Zubiaga et al., 2018). Prior work has explored Bangla fake news detection using linguistic and psychological features (Mahmud et al., 2026), but this targets fake news rather than rumours, is not domain-stratified, and does not characterise cross-domain linguistic variation; no domain-stratified Bangla rumour corpus exists, and no theoretical framework has examined the limits of surface-level detection in this language.

We introduce **MisInfo-11K** through four contributions: (i) the first domain-stratified Bangla rumour corpus spanning seven topical domains, establishing the empirical foundation for cross-domain analysis; (ii) 14 validated linguistic markers producing domain-specific deception fingerprints that reveal what properties distinguish Bangla rumours from authentic content, via χ^2 and Kruskal-Wallis analysis; (iii) the **Marker Signal Void (MSV)**, a novel chi-square-weighted scoring framework quantifying, for our validated 14-marker set, the performance ceiling of surface-level detection and identifying which specific instances fall outside its reach; (iv) a benchmark of three pretrained transformers and five traditional models quantifying whether contextual representations contribute signal beyond surface markers — MuRIL achieves Macro-F1 0.9800 and a marker-augmented Random Forest reaches 0.9750, surpassing XLM-RoBERTa.

2 Related Work

Rumour Detection. PHEME (Zubiaga et al., 2016) and RumourEval (Gorrell et al., 2019) established rumour detection on Twitter, covering breaking-news events and formalising stance/veracity subtasks, while Bi-GCN (Bian

et al., 2020) showed propagation structure carries signal beyond text. A consistent finding is that linguistic features degrade across event domains (Zubiaga et al., 2018), directly motivating our domain-stratified analysis.

Bangla Fake News Detection. Hossain et al. (2020) introduced BanFakeNews ($\approx 50K$), the foundational Bangla benchmark, with follow-up work applying CNN-LSTM, BiGRU, and BERT-based classifiers; all prior work is confined to fake news classification on a domain-undifferentiated corpus, with none addressing rumour detection, domain stratification, or detection limits.

Linguistic Markers and Low-Resource Misinformation. Claim verbs, attribution patterns, and negation density discriminate misinformation across languages (Newman et al., 2003), but analyses remain single-domain, masking cross-domain variability; low-resource languages remain severely underserved (Chalehchaleh et al., 2024), with deception cues transferring unreliably across morphologically distinct languages, reinforcing the need for language-specific corpus construction.

Pretrained Language Models for Bangla. Bhattacharjee et al. (2022) introduced BanglaBERT (27.5GB Bangla web text, state-of-the-art on four Bangla NLU tasks); MuRIL (Khanuja et al., 2021) extends multilingual BERT with dedicated support for 17 South Asian languages including Bangla, while XLM-RoBERTa (Conneau et al., 2020) is the standard high-capacity multilingual baseline. We evaluate all three on the first domain-stratified Bangla rumour corpus for reproducible cross-family baselines.

3 Dataset: MisInfo-11K

3.1 Collection Methodology

Rumour instances were collected from three dedicated Bangla fact-checking portals — **Rumor Scanner**,¹ **Boom Bangladesh**,² and **Jachai**³ — via a multi-threaded, rate-limited scraper. URL-level deduplication within the training split prevented duplicate collection. Each fact-check headline is treated as a rumour instance (label 1).

¹<https://rumorsscanner.com/>

²<https://boombd.com>

³<https://jachai.org>

We use only the fact-check portal’s headline, not the article body, for two reasons. First, the headline reproduces the claim as it actually circulated on social media — this is the unit that misleads readers, and fact-check portals preserve it verbatim as the object of their verification. The article body, by contrast, is the fact-checker’s own debunking explanation and would trivially leak the rumour label if included. Second, using headlines on both sides keeps the comparison at a consistent structural level: non-rumour instances are likewise drawn from news headlines, since headline text drives engagement and exposure on social media, independent of the outlet’s editorial register. The task is therefore defined at the level readers actually encounter misinformation headline against headline rather than full-article content.

Non-rumour instances were collected from nine established Bangladeshi news outlets **Naya Diganta**, **NTV**, **MY TV**, **Manobzamin**, **RisingBD**, **BD Pratidin**, **Bangla Vision**, **Kaler Kantho**, and **Jugantor** — via RSS feeds and paginated HTML listings, filtered by a Unicode Bangla character check and a 15–300 character length constraint. Non-rumour labels were assigned programmatically and never sourced from fact-check platforms to prevent label leakage.

3.2 Domain Taxonomy

Seven domains were defined a priori based on the thematic coverage of Bangla fact-checking portals: political, health, religious, cultural, sports, celebrity, and international. Labels were assigned automatically via a Bangla keyword lexicon curated per category. Rumour instances may carry multiple domain labels reflecting co-occurring themes; non-rumour instances carry exactly one label derived from the source category. Instances with no keyword match default to political, reflecting political content’s empirical dominance in Bangla misinformation and its broad topical overlap with unlabelled instances. We acknowledge this fallback introduces a structural asymmetry: the political domain’s marker profile is a mixture of genuinely keyword-matched content and residual unclassified instances. However, we note that this pooling effect is directionally conservative rather than inflationary — merging heterogeneous, topically diffuse content into a single bucket tends to dilute rather than manufacture a distinctive marker signature, since unrelated instances regress the do-

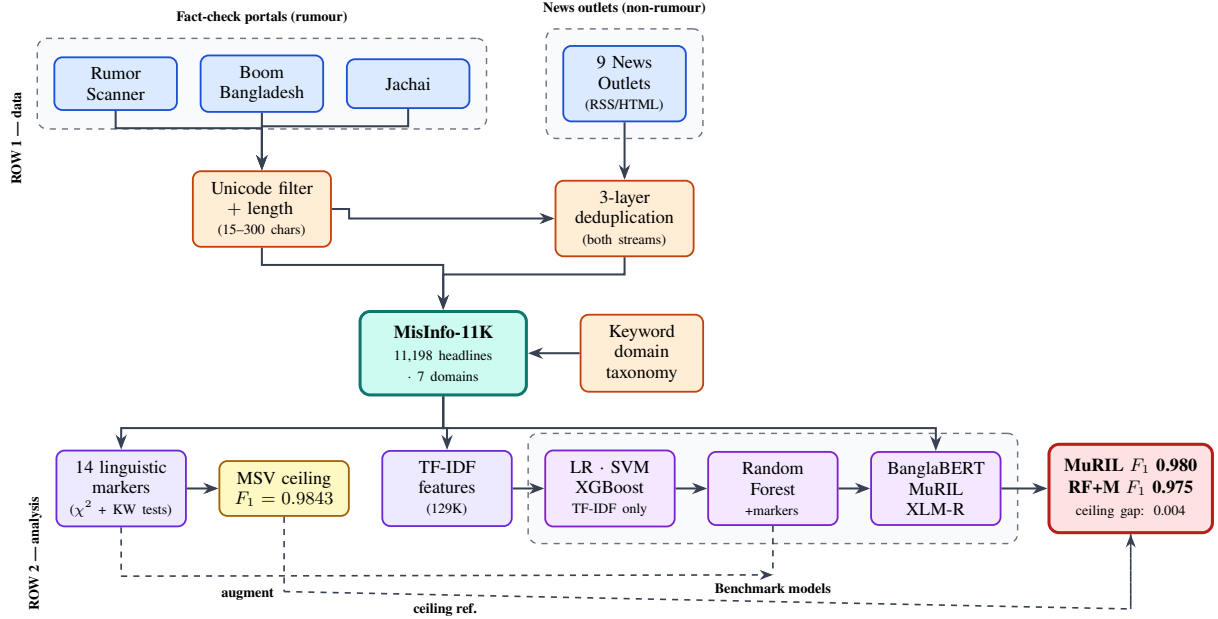


Figure 1: **MisInfo-11K pipeline.** **Row 1:** Rumour headlines from three fact-checking portals and non-rumour headlines from nine news outlets pass through quality-control filters and 3-layer deduplication (both streams) into the domain-stratified corpus. **Row 2:** Three parallel paths — (i) linguistic markers $\rightarrow$ MSV ceiling $F_1=0.9843$; (ii) TF-IDF $\rightarrow$ traditional classifiers with marker augmentation for Random Forest; (iii) transformer fine-tuning — converging on the benchmark results.

main’s mean activation rates toward the corpus-wide baseline. The Kruskal–Wallis result reported in Section 4.2 (13 of 14 markers significant across domains, $p < 0.05$) should therefore be read as, if anything, an underestimate of true cross-domain variation, since any homogenising effect of the fallback would work against detecting domain-specific signal rather than artificially creating it. We flag this as a taxonomy limitation in Section 7 (Limitations) and recommend that future work either introduce an explicit *Uncategorised* category or validate fallback assignments against a larger human-annotated sample.

3.3 Dataset Statistics

MisInfo-11K contains 11,198 instances across training (8,198), validation (1,400), and test (1,400) splits, each evaluation split balanced 50/50 between classes. Table 1 reports the full breakdown by split. Figure 2 shows domain distribution across both classes, revealing a notable asymmetry: political is the dominant rumour domain (2,407 instances) yet the most underrepresented in non-rumour (204), while sports and religious show the inverse pattern. This asymmetry motivates the domain-stratified analysis in Section 4.2.

3.4 Quality Control

Three layers of quality control were applied: exact-match deduplication on the text field to remove conflicting labels from headlines appearing across multiple sources; a URL and headline blacklist built from the training split to prevent any training instance from appearing in validation or test; and explicit cross-contamination checks confirming zero overlap across all three splits. Instances shorter than five characters, containing only ASCII, or matching known portal homepage titles were discarded.

Beyond deduplication, taxonomy reliability was validated through human annotation. Three annotators independently labelled a 200-instance stratified subset across all seven domains. Pairwise Cohen’s κ was computed for each domain, yielding a macro-averaged κ of 0.746 (*good* agreement; Landis and Koch 1977). Agreement was strongest for health ($\kappa = 0.861$) and sports ($\kappa = 0.847$), domains with unambiguous topical boundaries. Gold-standard labels for this subset were assigned by majority vote across the three annotators. To verify that rumour headlines reflect the circulating claim rather than the fact-checker’s own framing, we manually spot-checked a random sample of rumour instances for explicit meta-

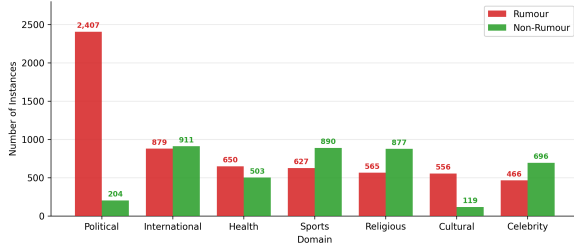


Figure 2: Domain distribution across rumour and non-rumour instances in the training set. Political is the dominant rumour domain (2,407 instances) yet the most underrepresented in non-rumour (204), revealing a structural asymmetry that motivates domain-stratified evaluation. Sports and religious show the inverse pattern.

linguistic markers (e.g., গুজব ‘rumour’, ভুয়া ‘fake’, সত্যতা যাচাই ‘fact-check verified’). We found such terms to be rare, supporting that the corpus predominantly captures the claim as circulated rather than the fact-checker’s verification framing.

Split	Total	Rumour	Non-R	Balanced
Train	8,198	3,998	4,200	✗
Validation	1,400	700	700	✓
Test	1,400	700	700	✓
Total	11,198	5,398	5,800	—

Table 1: MisInfo-11K dataset statistics by split. Non-R = Non-Rumour. The training split retains natural class imbalance (3,998 rumour vs. 4,200 non-rumour, 51.2% non-rumour); validation and test are balanced 50/50 to avoid confounding evaluation metrics with class prevalence. Training-set imbalance is addressed via class-weighted loss and a weighted random sampler (Section 5.2).

4 Linguistic Marker Framework

4.1 Marker Design and Validation

We design 14 linguistically motivated markers spanning four categories: *structural* (text length, word count, short title, long title, ends with question mark), *lexical* (claim verb, negation words, clickbait), *attribution* (has quote, unnamed source, named outlet), and *content* (has numbers, religious alarm, health claim). Markers were defined using Bangla-specific regular expressions validated against native speaker intuition and prior cross-lingual deception detection literature (Mahyoob et al., 2021; Zhou and Zafarani, 2020). Each marker was validated on the full combined dataset using chi-square tests for binary markers and Mann–Whitney U tests for continuous markers ($p < 0.0001$ for all 14). Table 2 reports discrimina-

tive weights $w_m = \chi_m^2 / \sum_{m'=1}^{14} \chi_{m'}^2$ and per-class activation rates. For the two continuous markers (word count and text length), the χ^2 column reports a variance-normalised effect size used solely for weight computation, while significance was measured using Mann–Whitney U tests.⁴

Figure 3 visualises binary marker activation rates and text length distribution by class.

Marker	χ^2	w_m	R	NR
Word count [†]	2881.07	0.311	9.83	7.30
Text length [†]	2494.79	0.269	65.80	49.55
Claim verb	1582.03	0.171	28.5%	1.6%
Long title	907.93	0.098	17.3%	0.9%
Has quote	287.90	0.031	5.4%	0.1%
Has numbers	254.17	0.027	10.2%	21.3%
Short title	237.01	0.026	1.2%	7.1%
Named outlet	151.66	0.016	2.8%	0.0%
Health claim	141.76	0.015	0.7%	4.2%
Negation words	87.59	0.009	16.0%	10.0%
Ends question	86.00	0.009	1.0%	3.8%
Unnamed source	64.43	0.007	1.8%	0.2%
Clickbait	50.87	0.006	1.4%	0.2%
Religious alarm	49.75	0.005	2.7%	0.9%

Table 2: Linguistic marker discriminative weights, ranked by χ^2 (chi-square statistic for binary markers; variance-normalised effect size for the two continuous markers[†] — see Section 4.1). $w_m = \chi_m^2 / \sum_{m'=1}^{14} \chi_{m'}^2$ is the normalised weight used in the WMS score (Eq. 1).

4.2 Domain Deception Fingerprints

We compute the mean activation rate of each marker separately for rumours in each of the seven domains, producing a 7×14 deception strategy profile matrix. Kruskal–Wallis tests confirm that 13 of 14 markers show statistically significant cross-domain variation ($p < 0.05$). Only negation_words fails to reach significance ($H = 10.34$, $p = 0.111$), rejecting the hypothesis that Bangla rumours employ a uniform linguistic strategy across topics.

Figure 4 shows three broad deception patterns by domain: *authority mimicry* (political/international, elevated claim_verb rates of 0.260/0.307) (Rashkin et al., 2017), *emotional provocation* (religious, highest religious_alarm rate at 0.196), and *fear amplification* (health, elevated health_claim/has_numbers co-occurrence) (Pérez-Rosas et al., 2018). Pairwise cosine similarities between domain profiles exceed 0.999 (range:

⁴Mann–Whitney U statistics: text length $U = 22,564,338$; word count $U = 23,117,884$; both $p < 0.0001$.

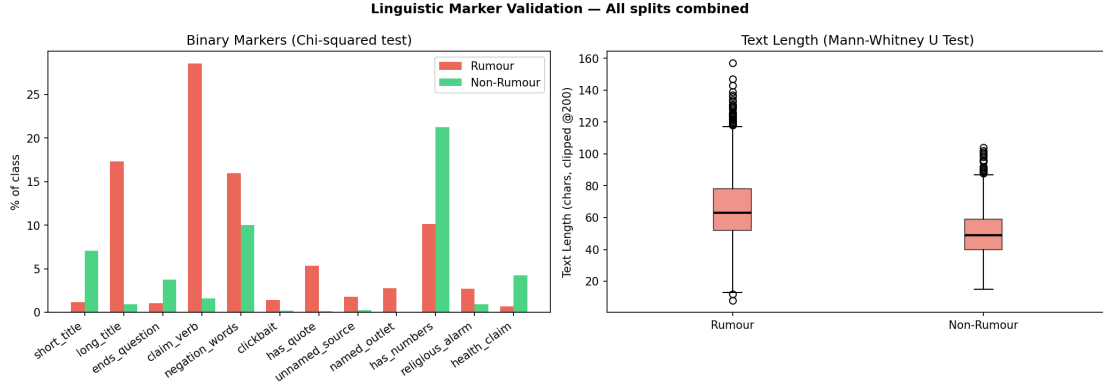


Figure 3: Left: binary marker activation rates by class (all χ^2 , $p < 0.0001$). Right: text length distribution by class (Mann–Whitney U, $p < 0.0001$). Rumours are systematically longer and more claim-verb dense, while non-rumours cite numbers and ask questions more frequently.

0.997–1.000),⁵ confirming that while domains differ in marker emphasis, they share a common underlying structure disrupted by domain-specific signals.

5 Experiments

5.1 Models

We benchmark eight models across two families. We restrict our benchmark to fine-tuned encoder models (BanglaBERT, MuRIL, XLM-RoBERTa) rather than large generative LLMs, since our focus is on establishing the surface-marker detection ceiling and quantifying the incremental value of contextual fine-tuning over hand-crafted features; zero-shot/few-shot LLM baselines are a natural extension we leave to future work.

Traditional Models. Five models combine character n-gram TF-IDF (2–5-grams) and word n-gram TF-IDF (1–3-grams), yielding 129,199 features:

1. Logistic Regression + TF-IDF
2. LinearSVC + TF-IDF
3. XGBoost + TF-IDF
4. Random Forest + TF-IDF + Markers
5. XGBoost + TF-IDF + Markers

Models 1–3 test surface pattern detection alone, while Models 4–5 test whether our 14 validated markers add discriminative signal beyond n-gram features.

Transformer Models. Three pretrained transformers are fine-tuned using a $[\text{CLS}] \rightarrow$

⁵Computed over the $72 = 21$ pairwise domain profile vectors.

Dropout(0.3) $\rightarrow$ Linear(768 $\rightarrow$ 2) classification head:

- **BanglaBERT** (Bhattacharjee et al., 2022), pretrained on 27.5 GB of Bangla text.
- **MuRIL** (Khanuja et al., 2021), a multilingual model with dedicated South Asian language support pretrained on 17 Indian languages including Bangla.
- **XLM-RoBERTa** (Conneau et al., 2020), the standard 125M-parameter multilingual baseline.

5.2 Training Setup

All transformers were trained for 4 epochs using AdamW ($lr = 2 \times 10^{-5}$, weight decay = 0.01), linear warmup over 10% of steps, maximum sequence length of 128 tokens, and batch size 16.

Class-weighted cross-entropy loss and a weighted random sampler jointly address training-set imbalance (3,998 rumour vs. 4,200 non-rumour instances). Best checkpoint selection used validation Macro-F1, and all random seeds were fixed at 42.

Traditional models used validation-set grid search over $C \in \{0.01, 0.1, 1.0, 10.0\}$ for Logistic Regression and SVM. XGBoost and Random Forest used fixed robust defaults (200 estimators, maximum depth 6).

5.3 Results

Table 3 reports Macro-F1, ROC-AUC, and accuracy on the held-out test set for all eight models.

MuRIL achieves the highest test Macro-F1 of 0.9800 (validation: 0.9821), followed by BanglaBERT (0.9771; validation: 0.9807) and XLM-RoBERTa (0.9736; validation: 0.9750). All

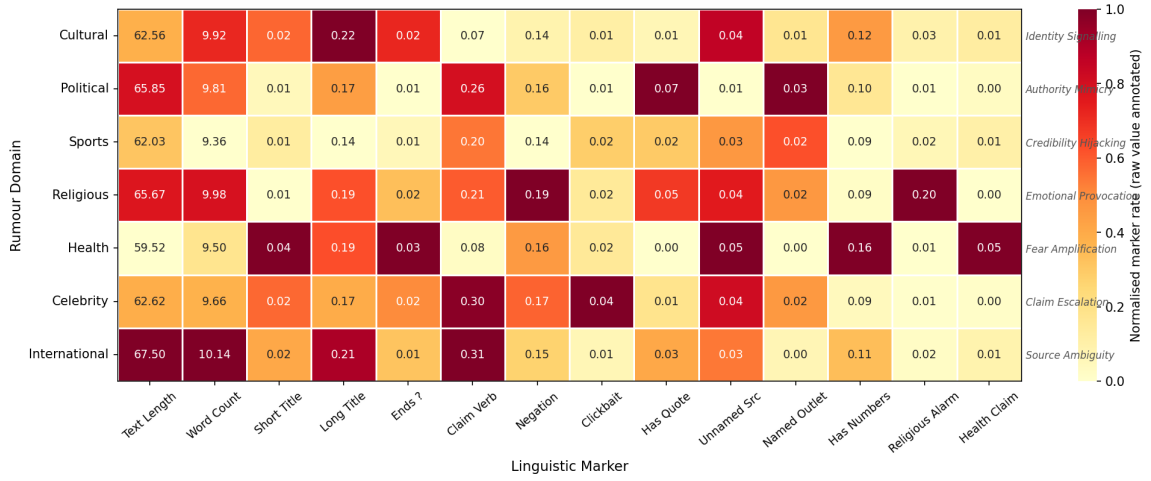


Figure 4: Domain deception strategy fingerprints. Colour intensity shows normalised marker activation rate, with raw values annotated. Strategy labels on the right axis reflect the dominant deception mechanism per domain.

validation-to-test Macro-F1 drops remain within 0.004 points (MuRIL: 0.0021; BanglaBERT: 0.0036; XLM-RoBERTa: 0.0014), indicating consistent generalisation.

Among traditional models, Random Forest augmented with our linguistic markers (0.9750) surpasses XLM-RoBERTa (0.9736), confirming hand-crafted markers carry genuine discriminative value beyond n-gram surface patterns. The +0.57 Macro-F1 gain of RF+Markers over the best TF-IDF-only model (LR and XGBoost, both 0.9693) quantifies the independent contribution of 14 validated markers.

Model	Macro-F1	ROC-AUC	Acc.
<i>Traditional Models</i>			
LR + TF-IDF	0.9693	0.9937	0.9693
SVM + TF-IDF	0.9643	0.9946	0.9643
XGBoost + TF-IDF	0.9693	0.9930	0.9693
RF + TF-IDF + M [†]	0.9750	0.9972	0.9750
XGB + TF-IDF + M [†]	0.9636	0.9934	0.9636
<i>Transformer Models</i>			
XLM-RoBERTa	0.9736	0.9937	0.9736
BanglaBERT	0.9771	0.9963	0.9771
MuRIL	0.9800	0.9971	0.9800

Table 3: Test-set results on the held-out test split ($n = 1,400$). Macro-F1 is macro-averaged F1 across the rumour and non-rumour classes; Acc. is overall accuracy. [†]M = augmented with our 14 linguistic markers. Best result per column in bold.

5.4 Cross-Dataset Generalisation

To assess transferability beyond Misinfo-11K, we evaluate our best-checkpoint BanglaBERT model

zero-shot on BanFakeNews (Hossain et al., 2020), a publicly available Bangla dataset comprising 7,000 authentic and 1,000 fake/satire news articles.

We sample 1,000 instances per class without any fine-tuning or domain adaptation, mapping BanFakeNews labels to our binary schema (authentic $\rightarrow$ non-rumour; fake/satire $\rightarrow$ rumour). We acknowledge this mapping is imperfect given our own argument that rumour and fake-news detection are categorically distinct tasks; we treat this transfer evaluation as a stress test of cross-task generalisation rather than a like-for-like benchmark. The model achieves Macro-F1 = 0.4728 and accuracy = 0.5325, with strong class asymmetry: recall on the fake class reaches 0.869, while authentic recall collapses to 0.196, producing 804 false positives against 131 false negatives.

This result is interpretable rather than anomalous, and arises from two compounding factors.

First, there is a severe input-length mismatch: MisInfo-11K consists of short claim headlines (mean 65.8 characters; truncated at 128 tokens), whereas BanFakeNews comprises full article bodies with mean length substantially exceeding this window. The model therefore operates on heavily truncated prefixes that fail to represent the article’s discriminative content.

Second, the two tasks differ categorically: rumour detection concerns unverified claims in active social-media circulation, while fake-news detection concerns fabricated article content with a different linguistic register, source structure, and propagation context (Zubiaga et al., 2018; Pérez-Rosas et al., 2018). These distinct feature distribu-

tions cannot be bridged through zero-shot transfer.

Rather than indicating a generalisation failure, this result provides direct empirical support for the core motivation of MisInfo-11K: Bangla rumour detection requires a dedicated headline-level corpus and task formulation that existing full-article fake-news resources cannot substitute.

6 Analysis

6.1 Marker Signal Void

We introduce the **Weighted Marker Score (WMS)** for an instance x :

$$\text{WMS}(x) = \sum_{m=1}^{14} w_m \cdot (f_m(x) - \mu_m^{\text{NR}}) \quad (1)$$

where $f_m(x)$ is the value of marker m , $w_m = \chi_m^2 / \sum_{m'=1}^{14} \chi_{m'}^2$ is its normalised discriminative weight (Table 2), and μ_m^{NR} is the mean of marker m over the non-rumour training split, computed without access to evaluation data to prevent leakage.

An instance lies in the **Marker Signal Void** when $\text{WMS}(x) \leq 0$: it carries no net positive marker signal relative to the non-rumour baseline, meaning surface-level linguistic cues are insufficient to distinguish it from authentic content.

While the ceiling formally bounds only the 14 markers validated here, these markers span all four linguistic categories established in cross-lingual deception-detection literature structural, lexical, attribution, and content-based cues (Mahyoob et al., 2021; Zhou and Zafarani, 2020) so the ceiling approximates, rather than exhaustively proves, the limit of surface-level detection more broadly. We do not claim exhaustiveness.

We find that 3.1% of test-set rumours (22 of 700) lie in the void. Figure 5 shows the WMS distribution by class: the two distributions overlap substantially near zero, confirming that a non-trivial subset of rumours is surface-linguistically indistinguishable from authentic content regardless of model capacity.

Examining the 22 void instances reveals three recurrent patterns:

1. *Meta-referential misinformation*: headlines from fact-checking organisations reporting their own annual detection statistics (e.g., “472 pieces of misinformation identified in March”), which carry the register of credible journalism.

2. *Government scheme impersonation*: rumours written in neutral administrative language with no claim verbs, vague attribution, or clickbait markers.
3. *Recurring death hoaxes*: bare declarative statements where the rumour keyword itself appears in both genuine fact-check headlines and fabricated claims, providing no discriminative signal.

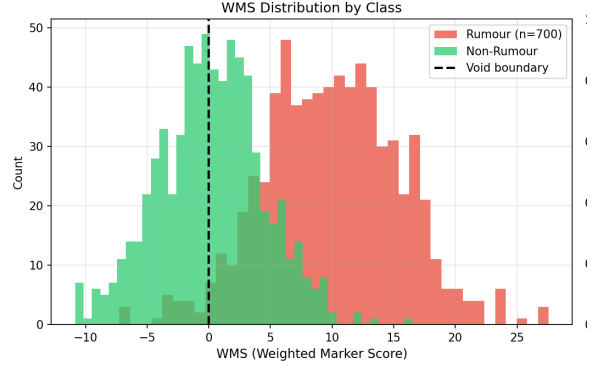


Figure 5: WMS distribution by class. The dashed line marks the void boundary ($\text{WMS} \leq 0$); 22 rumours (3.1%) fall below it. Overlap near zero confirms a subset of rumours is linguistically indistinguishable from authentic content at the surface level.

6.2 Detection Ceiling and Transformer Contribution

The 3.1% void rate establishes a hard theoretical ceiling on any system relying exclusively on the 14 surface markers. A perfect oracle achieves at most $\text{Macro-F1}_{\text{ceil}} = 0.9843$, corresponding to a theoretical recall ceiling of 0.9686 on the rumour class. All eight models operate below this ceiling. The ordering

TF-IDF only < TF-IDF + Markers < Transformer

holds consistently across the benchmark.

Explicit linguistic markers add +0.57 Macro-F1 points over distributional n-gram features (RF+Markers: 0.9750 vs. LR/XGB: 0.9693), while transformer pre-training adds a further +0.50 points over the best marker-augmented model (MuRIL: 0.9800 vs. RF+Markers: 0.9750).

Each component contributes independent, additive discriminative value.

MuRIL (FN = 10) recovers 12 of the 22 void rumours through contextual semantic representations; BanglaBERT (FN = 11) recovers 11; XLM-RoBERTa (FN = 14) recovers 8.

The 8–14 false negatives that persist across all three transformers constitute the irreducible difficulty of this dataset under text-only approaches (Zubiaga et al., 2018): rumours whose surface form is indistinguishable from authentic content even to contextual models.

6.3 Error Analysis

Table 4 summarises false positive and false negative counts for all three transformers on the test set.

MuRIL achieves the most favourable error profile (FN = 10, FP = 18), followed by BanglaBERT (FN = 11, FP = 21) and XLM-RoBERTa (FN = 14, FP = 23).

Model	Class	P	R	F1	FP	FN
MuRIL	Non-Rumour	0.9855	0.9743	0.9799	18	–
	Rumour	0.9746	0.9857	0.9801	–	10
BanglaBERT	Non-Rumour	0.9841	0.9700	0.9770	21	–
	Rumour	0.9704	0.9843	0.9773	–	11
XLM-RoBERTa	Non-Rumour	0.9797	0.9671	0.9734	23	–
	Rumour	0.9676	0.9800	0.9737	–	14

Table 4: Per-class results for transformer models ($n = 1,400$). P, R, and F1 denote precision, recall, and F1-score per class; FP and FN denote false positives and false negatives, reported once per model since each error is shared between the two classes (a false positive for Rumour is a false negative for Non-Rumour and vice versa).

False Negatives. Two categories dominate the errors shared by all three models: *meta-referential content* (Rumor Scanner/Jachai headlines reporting their own detection statistics, carrying the register of credible journalism and lying in the Marker Signal Void, $WMS \leq 0$), and *neutral-register impersonation* (government scheme rumours in bureaucratic language that activate no positive marker). Both expose a structural limitation: once a rumour adopts institutional or fact-checking register, neither marker-based nor distributional approaches can reliably flag it (Rashkin et al., 2017).

False Positives. The dominant pattern is *legitimate attributed speech* activating `claim_verb` ($w = 0.171$), where political/ international headlines quoting named parties are surface-identical to rumours quoting false claims; a second cluster is celebrity/ entertainment headlines whose dramatic vocabulary activates multiple lower-weight markers. This aligns with prior findings that epistemic markers misfire on legitimate attributed political

speech (Pérez-Rosas et al., 2018). Both error directions reveal consistent failure patterns across all three models.

7 Conclusion

We presented **MisInfo-11K**, the first domain-stratified Bangla rumour corpus (11,198 headlines, 7 domains), with an eight-model benchmark spanning traditional and transformer architectures. Our 14 validated linguistic markers (all $p < 0.0001$) reveal domain-specific deception fingerprints authority mimicry (political/international), emotional provocation (religious), and fear amplification (health) — with a +0.57 Macro-F1 gain from marker augmentation confirming their independent discriminative value. The Marker Signal Void analysis establishes Macro-F1 = 0.9843 as the ceiling for this marker set; MuRIL achieves 0.9800 (ROC-AUC = 0.9971), recovering 8–12 of the 22 void rumours through contextual representations. Zero-shot evaluation on BanFakeNews (Hossain et al., 2020) confirms rumour detection requires a dedicated headline-level resource. Future work includes propagation-graph integration (Bian et al., 2020), multimodal evidence fusion, and cross-lingual transfer to other South Asian languages.

Limitations

Domain and Temporal Scope. MisInfo-11K covers headlines from 2024–2026; models may not generalise to earlier periods or future data as misinformation strategies evolve.

Headline-Only Input. All models operate on headline text truncated to 128 tokens. The zero-shot collapse on BanFakeNews (Macro-F1 = 0.4728) confirms that article-level discriminative signals metadata, propagation context, and multimedia are absent from the current framework.

Formal Written Bangla. The corpus covers formal registers only. Informal social-media language dialectal variation, code-switching, WhatsApp forwards — is not represented, which may limit robustness on real-world misinformation content.

Taxonomy Reliability. Domain labels were assigned programmatically; only a 200-instance subset received human IAA validation ($\kappa = 0.746$). Domain labels are assigned via automatic keyword matching, with unmatched instances defaulting to political and rumour instances permitted multiple

domain labels while non-rumour instances receive exactly one; this asymmetry in label cardinality means the domain-level deception fingerprints in Section 4.2 should be read as indicative patterns rather than precise per-domain estimates.

Class Balance. We did not run a class-balanced downsampling ablation on the training split; given the modest imbalance (3,998 vs. 4,200) and our use of class-weighted loss / weighted sampling, we expect the effect to be small, but we flag this as an open empirical question.

Single-Language Scope. The marker framework is validated on Bangla only. Transfer to related South Asian languages (Hindi, Assamese, Sylheti) remains an open empirical question.

Data Availability. We will release MisInfo-11K including the train/validation/test splits, domain labels, and the 14 linguistic marker annotations — on GitHub upon publication to support reproducibility and further research on Bangla rumour detection.

References

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